

Extensions compatibility tests V2

This is the second (and hopefully successful) iteration of the compatibility tests for extensions. The goal is to have at least one "**stable**" (non-mutable) extension with the current **Extension Points** and **Shell API** points added to the code so that we can assert that everything is working fine in a multitude of Rancher versions.

Where to find extension code

<https://github.com/aalves08/elemental-ui/tree/compatibility-tests-version>

Architecture

- Idea is to create a nightly run test suite that should use the latest code in master as the basis for the shell package. Example of this is the **check-plugins** workflow, where there's a [bash script](#) that does the publishing of the shell package to a local instance of verdaccio and uses that to test builds of various extensions
- We should grab the built artifact of the extension with the "**latest**" shell and have it available to serve it locally (like in an extension we do **yarn serve-pkgs**) so that we can do a developer load
- We should test the extension in the latest patch version of Rancher for each minor version. We start on 2.10 and go up until **master** (at this point in time is tracking 2.15). Example would be to test 2.10.xxx, 2.11.xxx, 2.12.xxx, 2.13.xxx, 2.14.xxx and **latest**. Images can be found <https://hub.docker.com/r/rancher/rancher/tags>
- We can employ a matrix-like system if we want the tests to run parallel, either like the first iteration of these compatibility tests as per <https://github.com/rancher/dashboard/blob/master/.github/workflows/extensions-compatibility-tests.yaml> OR <https://github.com/rancher/dashboard/blob/master/.github/workflows/test.yaml> (use them as examples, take it with a grain of salt as implementation is not straightforward)
- We use Playright MCP server to handle the tests, for which I've added the first lot of test cases as per next session
- When we spin up a new Rancher system and enter the UI, do the Overall Setup step in the test cases
- run test cases

Test cases

Relevant links:

<https://extensions.rancher.io/extensions/next/api/overview>

<https://extensions.rancher.io/extensions/next/shell-api>

Overall setup (Prepare system to run, much like we do in Cypress tests)

- go to Header, click on the user avatar, then click the "Preferences" option
- under "Advanced features" tick the option "Enable Extension developer features"
- go to Extension main area in the side bar menu
- click on 3 dots menu, select the option "Developer Load"
- Paste the url that came from **yarn serve-pkgs** onto "Extension url" field, click load
- When extension is loaded (notification), click on the refresh/reload button

Extension point - ActionLocation.HEADER 1 (from rancher v2.7.2 onwards)

- Go to home page in the side bar menu
- Check that in header there's an element with data-testid="extension-header-action-action-one"
- Click on it should log to console "action executed 1"
- Using keyboard shortcut CMD+M should log to console "action executed 1"

Extension point - ActionLocation.HEADER 2 (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Check that in header there's an element with data-testid="extension-header-action-action-two"
- Click on it should log to console "action executed 2"
- Using keyboard shortcut CMD+B should log to console "action executed 2"

Extension point - Tab RESOURCE_DETAIL_PAGE (from rancher v2.12.6, v2.13.2, v2.14.0 onwards)

- Go to local cluster in the side bar menu
- Go to Service Discovery
- Go to Services list page
- Click on any given service to check the details page
- A new tab should appear with an element with data-testid="btn-detail-page-id" and aria-label="detail-page-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - Tab RESOURCE_CREATE_PAGE (from rancher v2.12.6, v2.13.2, v2.14.0 onwards)

- Go to local cluster in the side bar menu
- Go to Service Discovery
- Go to Services list page
- Click on "create" to create a new service
- Select "Cluster IP"
- A new tab should appear with an element with data-testid="btn-create-page-id" and aria-label="create-page-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - Tab RESOURCE_EDIT_PAGE (from rancher v2.12.6, v2.13.2, v2.14.0 onwards)

- Go to local cluster in the side bar menu
- Go to Service Discovery
- Go to Services list page
- On any service listed, go to the table row actions and click "Edit Config"
- A new tab should appear with an element with data-testid="btn-edit-page-id" and aria-label="edit-page-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - Tab RESOURCE_SHOW_CONFIGURATION (from rancher v2.12.6, v2.13.2, v2.14.0 onwards)

- Go to local cluster in the side bar menu
- Go to Service Discovery
- Go to Services list page
- Click on any given service to check the details page
- Click on "Show Configuration"
- A new tab should appear with an element with data-testid="btn-show-configuration-id" and aria-label="show-configuration-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - Tab CLUSTER_CREATE_RKE2 (from rancher v2.13.0 onwards)

- Go to "Cluster Management" in the side bar menu
- Click create
- Select custom cluster
- A new tab should appear with an element with data-testid="tab-cluster-create-rke2-id" and aria-label="cluster-create-rke2-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - Tab RESOURCE_DETAIL (legacy - up until rancher v2.14.0)

- Go to local cluster in the side bar menu
- Go to Workloads
- Go to Pods list page
- Click on any given pod to check the details page
- A new tab should appear with an element with data-testid="btn-pod-detail-id" and aria-label="pod-detail-label"
- Click on that tab, new tab content should appear with text "THIS IS A DEMO TAB"

Extension point - ActionLocation.TABLE (not bulkable) (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Apps
- Go to Repositories list page
- On any Repo listed, go to the table row actions and check that there's a table action called "Demo table action"
- Click on it should log to console "table action executed 1"
- On any Repo listed, go to the table row actions and check that there's a table action called "Demo bulkable action"
- Click on it should log to console "table action executed 2"

Extension point - ActionLocation.TABLE (bulkable) (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Apps
- Go to Repositories list page
- Select a couple of Repos using the checkbox on the left of them
- There should be appearing above the list an enabled button with the text "Demo bulkable action"
- Click on it should log to console "table action executed 2"

Extension point - PanelLocation.RESOURCE_LIST (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Apps
- Go to Repositories list page
- There should be a banner being displayed with the text "Just a sample banner to show that we can render anything here"

Extension point - PanelLocation.DETAILS_MASTHEAD & PanelLocation.DETAILS_TOP - 1 (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Apps
- Go to Repositories list page
- Click on any Repo to see the details page
- There should be a banner being displayed with the text "This is a generic masthead component example"
- There should be a banner being displayed with the text "This is an example on DetailTop"

Extension point - PanelLocation.DETAILS_MASTHEAD & PanelLocation.DETAILS_TOP - 2 (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Apps
- Go to Repositories list page
- On any Repo listed, go to the table row actions and click "Edit Config"
- There should be a banner being displayed with the text "This is a generic masthead component example"
- There should be a banner being displayed with the text "This is another component example for masthead details - edit view"

Extension point - PanelLocation.ABOUT_TOP (from rancher v2.13.0 onwards)

- Go to the About page from the side bar menu (click on version number on bottom of side bar)
- There should be a banner being displayed with the text "Just a sample banner to show that we can render anything here"

Extension point - CLUSTER_DASHBOARD_CARD (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Check that there's a new card on the cluster explorer page with the text "Demo card title 1"

Extension point - Table Hook (from rancher v2.14.0 onwards)

- Go to local cluster in the side bar menu
- Go to Workloads
- Go to Pods list page
- A console log should appear with the text "TABLE HOOK TRIGGERED"
- Applying a filter to the Pods list view should make appear a console log with the text "TABLE HOOK TRIGGERED"
- Applying sorting to any table col to the Pods list view should make appear a console log with the text "TABLE HOOK TRIGGERED"

Extension point - Add table column 1 - custom formatter (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Storage
- Go to Secrets list page
- There should be a new table column with the label "Extension Col - Example 1" and the table cell value should contain the text "Formatter: Custom Cell Value 1"

Extension point - Add table column 2 - pagination (from rancher v2.7.2 onwards)

- Go to local cluster in the side bar menu
- Go to Storage
- Go to Configmaps list page
- There should be a new table column with the label "Extension Col - Example 2" and the table cell value should contain the text "Custom Cell Value 2" OR the name of the resource (same as "name" col value)

Shell API - Slidein API test (from rancher v2.14.0 onwards)

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "shell-api-demo"
- Click on "Test Slide-in API"
- A new slide-in panel should appear with the title "Hello from SlideIn panel!"

Shell API - Modal API test (from rancher v2.14.0 onwards)

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "shell-api-demo"
- Click on "Test Modal API"
- A new modal should appear with a title "Sample general title"
- A new modal should appear with a button "Cancel"
- A new modal should appear with a button "Add"

Shell API - Notification API test (from rancher v2.14.0 onwards)

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "shell-api-demo"
- Click on "Test Notification API"
- A new notification should appear with a title "Some notification title"
- A new notification should appear with a text body "Hello world! Success!"

Shell API - System API test (from rancher v2.14.0 onwards)

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "shell-api-demo"
- Click on "Test System API"
- The box under "System API Data:" should switch text from "No System Info yet. Press the button to fetch it." to system information with the following keywords: gitCommit, isDevBuild, isPrereleaseVersion, isRancherPrime, kubernetesVersion, rancherVersion

Elemental Extension - setup

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "Dashboard"
- Click on the button "Install Elemental Operator"
- Click on the button "Next"
- Click on the button "Install"
- Wait for the backend part to be successfully installed (terminal should contain "SUCCESS: helm upgrade --install=true --labels=catalog.cattle.io /cluster-repo-name=rancher-charts" and should appear "disconnected"
- Close terminal
- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "Dashboard"
- Page should have title "OS Management Dashboard"

Elemental Extension - EDIT/CREATE interface

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "Registration Endpoint"
- Click "Create"
- Fill the "name" field with "demo-reg-endpoint-1"
- Click "create"
- A new page should appear (details page) with the title/text "Registration Endpoint: demo-reg-endpoint-1" (careful that it may be two different HTML elements)
- Click on the sub-menu entry called "Registration Endpoint"
- There should be a new table entry with name "demo-reg-endpoint-1"
- Click on the sub-menu entry called "Dashboard"
- There should be a new table entry with name "demo-reg-endpoint-1" on the table "Registration Endpoints" in the dashboard view

Elemental Extension - EDIT/CREATE YAML interface

- Go to the Elemental extension in the side bar menu
- Click on the sub-menu entry called "Inventory of Machines"
- Click "Create from YAML"
- Replace "#string" in [metadata.name](#) for "demo-mach-inv-1"
- Click "create"
- There should be a new table entry with name "demo-mach-inv-1"

